

SPCE 5065
FINAL
SUMMER 2026

I completed this assignment using only authorized references. These are the textbook, your personal notes, any material on the course Canvas page, and a calculator.

Signed: _____

Printed: _____

1. _____ /10

2. _____ /10

3. _____ /10

4. _____ /8

5. _____ /8

6. _____ /10

7. _____ /9

8. _____ /10

9. _____ /15

10. _____ /10

Bonus: ____/5

SPCE 5065

FINAL

SUMMER 2026

1. (10 pts.) True/False

- a. _____ A spacecraft coated with a thermal material with a high emissivity will have a lower equilibrium temperature than material with a low emissivity.
- b. _____ Safe mode is often used to recover a spacecraft from a Micrometeoroid or Orbital Debris (MMOD) impact.
- c. _____ Free fall and zero gravity are the same.
- d. _____ For a spacecraft in GEO it is best to use a large positive bias voltage.
- e. _____ REMs and RADs are equivalent in the Van Allen Radiation belts.
- f. _____ The neutral environment, as explained in this course, is the same as the vacuum environment.
- g. _____ The most common contamination cleanliness level is class 10,000, which is equivalent to ISO level 7.
- h. _____ Galactic cosmic rays generally consist of higher-energy particles than those produced during solar particle events.
- i. _____ A Single Event Upset (SEU) permanently damages a microelectronic device and typically requires replacing the component.
- j. _____ A 1-mm debris particle in LEO is generally considered too small to cause significant spacecraft damage.

SPCE 5065

FINAL

SUMMER 2026

2. (10 pts)

I. What is the relevance of a Single-Event Latch-up to a spacecraft's performance during a space mission?

- a. A latch-up can significantly impact spacecraft performance by causing excessive current draw, requiring protective power shutdown or reset, and potentially leading to permanent damage if not mitigated.
- b. A latch-up only temporarily impacts system performance because spacecraft fault-tolerance architectures always detect the event, isolate the affected circuit, and restore normal operation without risk of permanent damage.
- c. A latch-up catastrophically impacts system performance by immediately destroying the affected electronics, making the spacecraft or subsystem permanently inoperable without any possibility of recovery.
- d. A latch-up primarily impacts system performance by increasing the susceptibility of spacecraft electronics to subsequent single-event effects, resulting in progressively higher upset rates throughout the mission.
- e. None of the above.

II. Outgassing is commonly due to which of the following (*select all that apply*)

- a. decantation
- b. decomposition
- c. deflagration
- d. diffusion
- e. desorption

III. Solar ultraviolet radiation alters which of the following, which results in temperature changes on the spacecraft? (choose the *best* answer)?

- a. strength to flexibility ratio
- b. reflectivity to transmissivity ratio
- c. absorptivity to emissivity ratio
- d. stress to strain ratio
- e. decomposition to desorption ratio

SPCE 5065

FINAL

SUMMER 2026

IV. The primary purpose of the thin front bumper in a Whipple shield is to

- a. Reflect the particle back into space.
- b. Vaporize or fragment the projectile before it reaches the rear wall.
- c. Absorb all of the impact energy by itself.
- d. Reduce spacecraft charging.
- e. Prevent thermal expansion.

V. The Half-Value Layer (HVL) of a material is defined as the thickness required to

- a. Reduce the radiation energy by one-half.
- b. Reduce the photon flux by one-half.
- d. Double the shielding effectiveness.
- d. Reduce the absorbed dose to zero.
- e. Eliminate secondary radiation.

3. (10 pts.) You are designing a *GEO satellite* with a sensor that will identify a target on earth's surface at 215 THz. Will this be an effective design? Explain your rationale. Hint: Consider both plasma oscillations (Lessons 7 & 8) and atmospheric transmittance (Lesson 2).

SPCE 5065
FINAL
SUMMER 2026

4. (8 pts.) Correlated the charged particle radiation flux density with solar activity by labelling “lower” or “higher.”

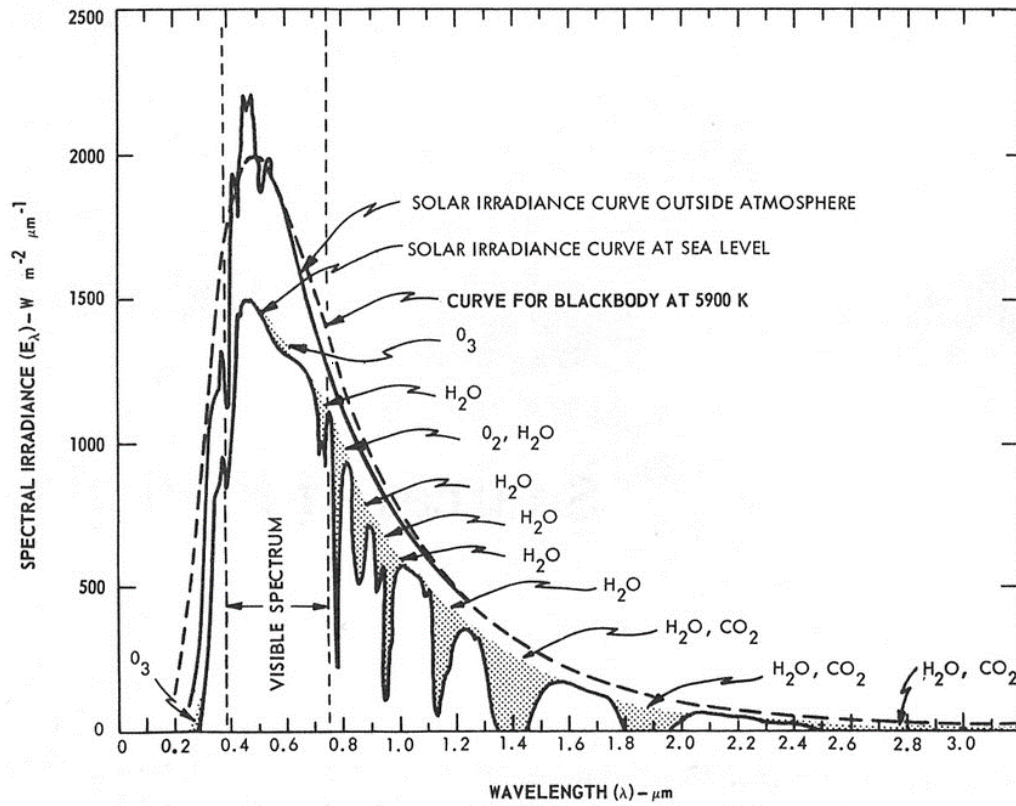
	Solar Min	Solar Max
Trapped radiation	-----	-----
- electrons		
- protons		
Galactic Cosmic Rays		
Solar Particle Events		

5. (8 pts) NO (Nitrous oxide) is a common element near earth. You may assume the bond energy is 1.67 eV in the lower ionosphere.

a. Identify the maximum wavelength a photon may have and still be capable of severing the bond.

b. Is this a significant part of the solar spectrum (i.e., > 50%)? You do not need to quantify your answer.

SPCE 5065
FINAL
SUMMER 2026

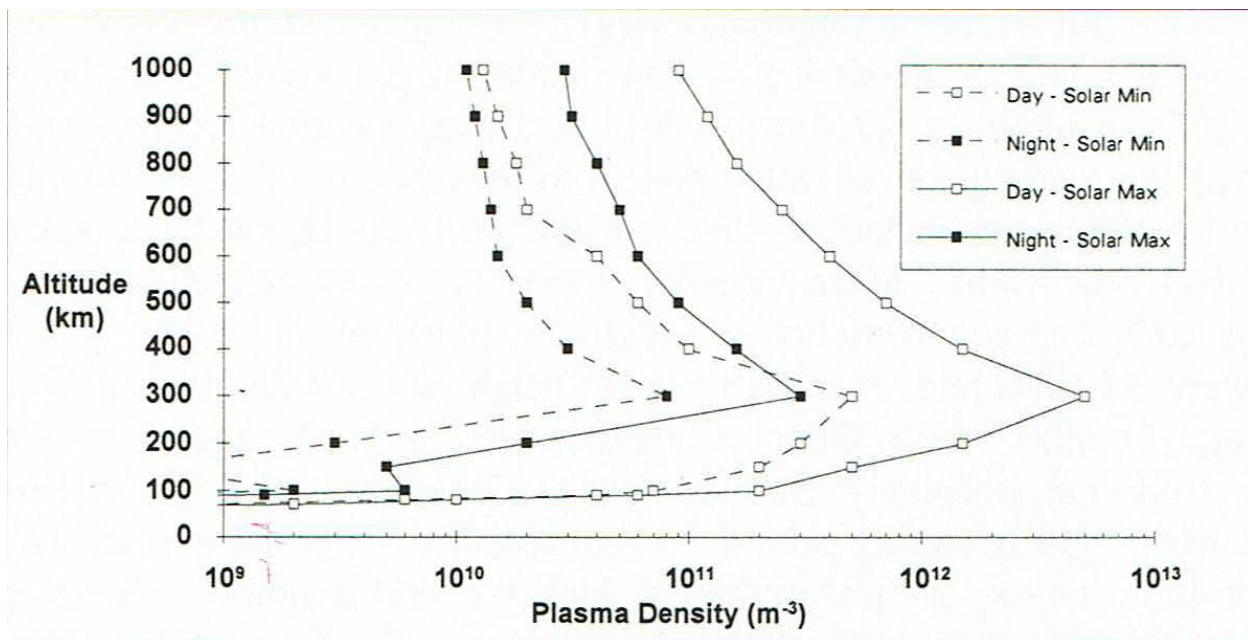


SPCE 5065

FINAL

SUMMER 2026

6. (10 pts) What is the approximate range of plasma frequencies (in MHz) associated with a 1000 km altitude orbit? Why are we interested in knowing that range of frequencies?



SPCE 5065
FINAL
SUMMER 2026

7. (9 pts.) A CubeSat is planned for a 550 km sun-synchronous orbit and has a limited mass budget. The mission requires:

- 5-year lifetime
- Reliable communications
- Low development cost
- Commercial off-the-shelf electronics whenever possible

Select **three** environmental hazards that you would address first. Rank them from most to least important and justify your ranking.

For each hazard, recommend one mitigation strategy and discuss one tradeoff introduced by your solution.

SPCE 5065
FINAL
SUMMER 2026

8. (10 pts). What is the expected worst-case excess range and time delay for a satellite operating at 500 km in K-band. State your assumptions and show your work.

FREQUENCY BANDS AVAILABLE FOR SATELLITE COMMUNICATION

Band	Frequency Range	Total Bandwidth	General Application
L	1 to 2 GHz	1 GHz	Mobile satellite service (MSS)
S	2 to 4 GHz	2 GHz	MSS, NASA, deep space research
C	4 to 8 GHz	4 GHz	Fixed satellite service (FSS)
X	8 to 12.5 GHz	4.5 GHz	FSS military, terrestrial earth exploration, and meteorological satellites
Ku	12.5 to 18 GHz	5.5 GHz	FSS, broadcast satellite service (BSS)
K	18 to 26.5 GHz	8.5 GHz	BSS, FSS
Ka	26.5 to 40 GHz	13.5 GHz	FSS

SPCE 5065
FINAL
SUMMER 2026

9. (15 pts.) A low-cost UCCS cube shaped satellite with each side 1 m^2 will operate in LEO at an altitude of 300 km. Assume an. ternal heat generation of 100 Watts and that only one side faces the sun and one side faces the Earth. Only black paint is used on all six sides. Black paint has an absorptivity of 0.975 and emissivity of 0.874. State any additional assumptions you make for this problem. There are no solar arrays on the satellite, so the satellite will use primary batteries that need to be kept between 0 and 15°C .

a. What are the equilibrium temperatures in the sun and in the eclipse?

b. Will this be adequate for your mission? Explain.

c. If you could change up to four sides (the sides NOT facing the sun or Earth), what solution would you recommend? Use the lowest cost solution you can find. Clearly state your final equilibrium temperatures and the cost associated with your solution.

SPCE 5065
FINAL
SUMMER 2026

10. (10 pts.) **Discuss** three ways satellites can be designed to better survive MMOD impacts.

SPCE 5065
FINAL
SUMMER 2026

BONUS (+5 pts.)

11. Derive an expression for the rate of change of a satellite's altitude, i.e., $\dot{R}$, due to an applied ΔV . Assume the satellite's orbit is circular and use a method similar to the method we used to derive $\dot{R}$ due to drag.